

Course Code	21CSE451T	Course Name	PATTERN RECOGNITION TECHNIQUES		Course Category	E	PROFESSIONAL ELECTIVE								L	T	P	C				
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Pre-requisite Courses		Nil	Co- requisite Courses		Progressive Courses		Nil															
Course Offering Department		School of Computing			Data Book / Codes / Standards		Nil															
Course Learning Rationale (CLR):			The purpose of learning this course is to:																			
CLR-1:	Know about various techniques in pattern recognition																					
CLR-2:	Develop good knowledge of Bayesian Theory																					
CLR-3:	Explore the various non-Parametric techniques																					
CLR-4:	Understand the Linear Discriminant functions																					
CLR-5:	Utilize Artificial Neural networks for classification problems																					
Course Outcomes (CO):			At the end of this course, learners will be able to:																			
CO-1:	Identify the various techniques involved in pattern recognition																PSO-3					
CO-2:	Summarize Bayesian decision theory and Bayesian learning																PSO-2					
CO-3:	Categorize the various non-Parametric techniques																PSO-1					
CO-4:	Formulate appropriate Linear Discriminant Function, Nearest neighbor rule, Neural Network and SVM for the problem statement																					
CO-5:	Demonstrate the Artificial Neural Network based Pattern recognition																					
Unit-1 - Introduction to Pattern recognition Systems																			9 Hour			
Basics of Probability, Independence of events, Conditional and Joint probability, Machine perception, Pattern Recognition Systems, The Design cycle, Learning and Adaptation, Minimum-error-rate classification, Classifiers, Discriminant functions, Decision surfaces, Normal density, and discriminant functions																						
Unit-2 - Parameter Estimation Methods																			9 Hour			
Maximum Likelihood Estimation, Bayesian Estimation, Bayesian Parameter Estimation: Gaussian case, Bayesian Parameter Estimation: General Theory, Problems of Dimensionality, Component Analysis and Discriminants, Expectation-Maximization, Hidden Markov Model																						
Unit-3 - Non-Parametric Techniques																			9 Hour			
Density Estimation, Parzen Windows, K- Nearest Neighbor Estimation, The Nearest Neighbor Rule, Metrics and Nearest Neighbor classification, Fuzzy classification, reduced coulomb energy networks, Approximations by series expansions																						
Unit-4 - Linear Discriminant Functions																			9 Hour			
Linear Discriminant Functions and Decision Surfaces, Generalized Linear Discriminant Functions, Two-category linearly separable case, Perceptron criterion functions, Relaxation procedures, Nonseparable Behavior, Minimum Squared Error procedures, Linear Programming Algorithms, Support Vector Machines																						
Unit-5: Multilayer Neural Networks and Nonmetric Methods																			9 Hour			
Introduction to Neural Networks. Multilayer Neural Networks: Feedforward operations and classification, Backpropagation algorithms, Nonmetric methods: Decision Trees, CART, Applications: Face recognition System																						

Learning Resources	1. R.O. Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley 2001 2. S. Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009 3. C.M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006 4. P.A Devijver and J. Kittler, Pattern Recognition: A Statistical Approach, Prentice-Hall, International, Englewood Cliffs, NJ, 1980	5. K. Fukunaga, Introduction to Statistical Pattern Recognition, 2nd Ed. Academic Press, New York, 1990. 6. Wu, Jianxin. Essentials of Pattern Recognition: An Accessible Approach. United Kingdom, Cambridge University Press, 2020. 7. Bengio, Y., Goodfellow, I., Courville, A. (2016). Deep Learning. United Kingdom: MIT Press.
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Learning Assessment		Continuous Learning Assessment (CLA)					
	Bloom's Level of Thinking	Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	15%	-	15%	-	15%	-
Level 2	Understand	25%	-	25%	-	25%	-
Level 3	Apply	30%	-	30%	-	30%	-
Level 4	Analyze	30%	-	30%	-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
Total		100 %		100 %		100 %	

Course Designers		Experts from Higher Technical Institutions		Internal Experts	
Experts from Industry					
1.	Dr.Peter PethuruRaj, Jio Cloud, Bengaluru				
2.	Mr. Srivathsa Pasumarthi, Senior Research Scientist Subtle Medical, Menlo Park, California, USA			1. Dr.B.Baranidharan, Associate Professor, Dept of C.Tech, SRMIST	